Quantification: 877_wt_z073c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
---------

| measure | value |
|---|---|
| lacuna count | 12 |
| interior lacuna count | 11 |
| frame-edge lacuna count | 1 |

Interior lacunae only: mean, median and sample SD (n = 11)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 2834.45 | 2671 | 1845.67 | px^2 |
| major_axis_length_px | lacuna shape | 93.8249 | 101.574 | 35.0777 | px |
| minor_axis_length_px | lacuna shape | 39.1988 | 36.8708 | 15.391 | px |
| aspect_ratio | lacuna shape | 2.5302 | 2.1489 | 1.062 | unitless |
| eccentricity | lacuna shape | 0.8781 | 0.8851 | 0.077 | unitless |
| solidity | lacuna shape | 0.7896 | 0.7983 | 0.0629 | unitless |
| roots_count | headline | 10.0909 | 9 | 7.8798 | count |
| ring_length_r30_px | headline | 374.273 | 352 | 238.754 | px |
| ring_length_r60_px | reported | 898.636 | 893 | 490.753 | px |
| owned_length_px | ownership-dependent | 1761.4 | 911.365 | 1768.04 | px |
| edge_count | ownership-dependent | 86.7273 | 40 | 92.2595 | count |
| mean_edge_length_px | ownership-dependent | 22.3691 | 20.3958 | 12.8627 | px |
| perimeter_px | normalised | 296.848 | 294.09 | 109.089 | px |
| ring_area_r30_px2 | normalised | 9683.27 | 9629 | 3127.42 | px^2 |
| ring_area_r60_px2 | normalised | 23306.1 | 24171 | 7222.91 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9872 | 1 | 0.0424 | unitless |
| in_frame_fraction_r60 | normalised | 0.9667 | 1 | 0.0792 | unitless |
| ring_density_r30 | normalised | 0.0350706 | 0.0374909 | 0.0155736 | px^-1 |
| ring_density_r60 | normalised | 0.0357389 | 0.0377538 | 0.0123656 | px^-1 |
| roots_per_100px_perimeter | normalised | 2.9461 | 3.4382 | 1.7349 | per 100 px |
| ring_attached_length_r30_px | network v2 | 325.727 | 310 | 242.128 | px |
| ring_attached_length_r60_px | network v2 | 685 | 538 | 543.546 | px |
| ring_length_w_r30_px | network v2 | 428.211 | 414.747 | 272.583 | px |
| ring_length_w_r60_px | network v2 | 1033.12 | 1019.55 | 566.555 | px |
| sholl_crossings_r10 | network v2 | 8.6364 | 8 | 5.1434 | count |
| sholl_crossings_r20 | network v2 | 10.9091 | 11 | 7.0916 | count |
| sholl_crossings_r30 | network v2 | 13.9091 | 16 | 7.2864 | count |
| loop_count | loops, report only | 8.9091 | 4 | 11.8275 | count |
| loop_edge_share | loops, report only | 0.2615 | 0.3172 | 0.2019 | share |
| ring_width_mean_r30_px | width | 6.5624 | 6.3758 | 0.9024 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.01666e+06 | px^2 |
| total_skeleton_length_px | 31448 | px |
| canalicular_length_density_per_px | 0.0309327 | px^-1 |
| junction_count | 669 | count |
| junction_density_per_px2 | 0.00065804 | px^-2 |
| lacuna_count | 12 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.18e-05 | px^-2 |
| skeleton_component_count | 368 | count |
| mean_component_length_px | 85.4565 | px |
| median_component_length_px | 24 | px |
| field_length_density_w_per_px | 0.0354722 | px^-1 |
| field_density_without_flagged_per_px | 0.0311809 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 498 | px |
| unattached_component_count | 27 | count |
| canalicular_length_density_attached_per_px | 0.0309327 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0314226 | px^-1 |
| loop_count | 122 | count |
| loop_edge_share | 0.2879 | share |
| edge_count | 1643 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.4661 | px |
| width_median_px | 8 | px |
| width_p10_px | 4 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 29992 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 954 | 58.9902 | 25.4792 | 2.3152 | 0.9019 | 0.6974 | 1.3025 | 49.6237 | 860.655 | False |
| 2 | 6236 | 133.392 | 66.6525 | 2.0013 | 0.8662 | 0.7565 | 0.7065 | 193.518 | 518.807 | False |
| 3 | 1740 | 108.483 | 22.6499 | 4.7895 | 0.978 | 0.7716 | 1.1221 | 244.43 | 785.516 | False |
| 4 | 4847 | 142.75 | 45.7551 | 3.1199 | 0.9472 | 0.8253 | 0.4966 | 381.669 | 319.923 | False |
| 5 | 2583 | 119.873 | 29.7486 | 4.0295 | 0.9687 | 0.7747 | 0.5429 | 394.401 | 41.6202 | False |
| 6 | 4698 | 104.479 | 62.6265 | 1.6683 | 0.8004 | 0.7983 | 0.887 | 422.75 | 620.834 | False |
| 7 | 2761 | 84.1795 | 44.2264 | 1.9034 | 0.8509 | 0.8577 | 1.3929 | 435.645 | 810.324 | False |
| 8 | 3441 | 99.3913 | 46.2527 | 2.1489 | 0.8851 | 0.8428 | 1.0959 | 709.919 | 822.946 | False |
| 9 | 2671 | 101.574 | 36.8708 | 2.7549 | 0.9318 | 0.8211 | 0.7949 | 796.647 | 571.938 | False |
| 10 | 654 | 43.6248 | 28.1155 | 1.5516 | 0.7646 | 0.6728 | 1.4274 | 853.878 | 586.749 | False |
| 11 | 594 | 35.3373 | 22.8099 | 1.5492 | 0.7638 | 0.8672 | 0.7727 | 907.293 | 697.131 | False |
| 12 | 740 | 61.1371 | 17.5677 | 3.4801 | 0.9578 | 0.8675 | -1.5683 | 1016.29 | 147.323 | True |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 9 | 351 | 769 | 2021.38 | 80 | 25.2673 | 6.043 |
| 2 | 25 | 793 | 1730 | 5467.12 | 277 | 19.7369 | 7.436 |
| 3 | 7 | 352 | 989 | 2005.23 | 101 | 19.8537 | 7.5283 |
| 4 | 19 | 518 | 1283 | 2957.39 | 145 | 20.3958 | 7.6229 |
| 5 | 5 | 227 | 592 | 528.375 | 28 | 18.8705 | 5.7804 |
| 6 | 19 | 732 | 1544 | 4074.78 | 222 | 18.3549 | 7.324 |
| 7 | 9 | 361 | 893 | 753.24 | 34 | 22.1541 | 7.269 |

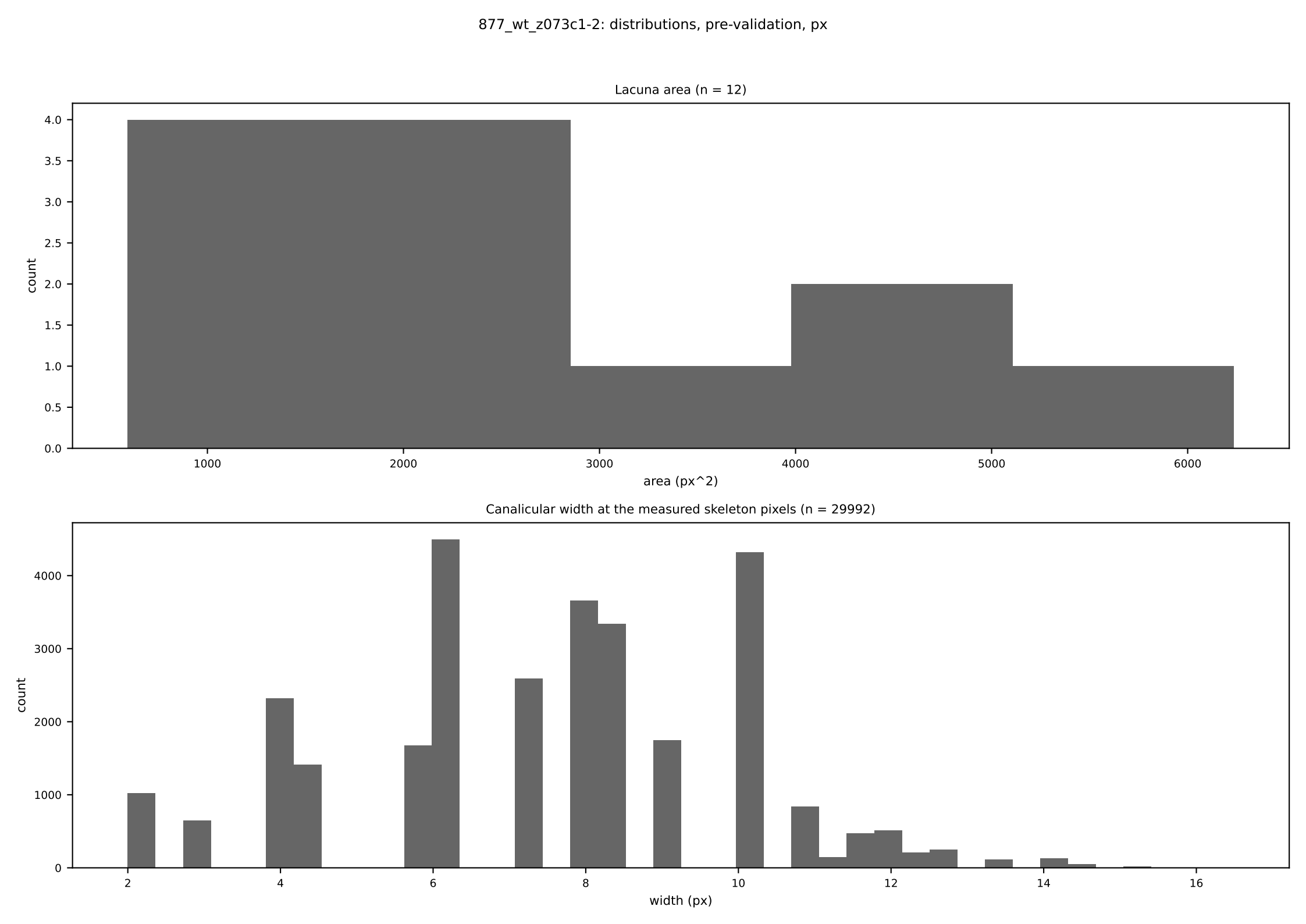
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 8 | 11 | 441 | 993 | 601.014 | 26 | 23.1159 | 6.212 |
| 9 | 6 | 207 | 689 | 911.365 | 40 | 22.7841 | 6.3758 |
| 10 | 1 | 32 | 134 | 55.5269 | 1 | 55.5269 | 5.3467 |
| 11 | 0 | 103 | 269 | 0 | 0 | 0 | 5.2483 |
| 12 | 0 | 0 | 0 | 0 | 0 | 0 | |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 218.244 | 7625 | 18201 | 1 | 0.8799 | 0.0460328 | 0.0422504 | 4.1238 |
| 2 | 510.192 | 15054 | 34735 | 1 | 1 | 0.052677 | 0.0498057 | 4.9001 |
| 3 | 294.09 | 10047 | 25475 | 1 | 1 | 0.0350353 | 0.0388224 | 2.3802 |
| 4 | 387.789 | 13352 | 32069 | 1 | 1 | 0.0387957 | 0.0400075 | 4.8996 |
| 5 | 312.576 | 9613 | 20973 | 0.8593 | 0.7542 | 0.0236139 | 0.0282268 | 1.5996 |
| 6 | 397.546 | 12204 | 28960 | 1 | 1 | 0.0599803 | 0.0533149 | 4.7793 |
| 7 | 261.764 | 9629 | 24171 | 1 | 1 | 0.0374909 | 0.0369451 | 3.4382 |
| 8 | 305.262 | 10522 | 26302 | 1 | 1 | 0.0419122 | 0.0377538 | 3.6035 |
| 9 | 288.948 | 8146 | 18856 | 1 | 1 | 0.0254112 | 0.0365401 | 2.0765 |
| 10 | 164.894 | 4521 | 10410 | 1 | 1 | 0.00707808 | 0.0128722 | 0.6064 |
| 11 | 124.024 | 5803 | 16215 | 1 | 1 | 0.0177494 | 0.0165896 | 0 |
| 12 | 159.918 | 3763 | 10281 | 0.5406 | 0.5269 | 0 | 0 | 0 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 310 | 538 | 406.09 | 897.477 | 8 | 8 | 16 |
| 2 | 768 | 1668 | 893.754 | 1984.24 | 15 | 23 | 19 |
| 3 | 324 | 895 | 415.889 | 1148.12 | 8 | 11 | 17 |
| 4 | 430 | 1163 | 594.571 | 1493.73 | 12 | 15 | 21 |
| 5 | 157 | 324 | 257.108 | 676.754 | 7 | 5 | 11 |
| 6 | 717 | 1377 | 841.311 | 1772.95 | 19 | 22 | 26 |
| 7 | 346 | 776 | 414.747 | 1019.55 | 10 | 11 | 13 |
| 8 | 307 | 484 | 503.274 | 1134.4 | 7 | 13 | 16 |
| 9 | 147 | 220 | 238.35 | 785.139 | 4 | 7 | 8 |
| 10 | 27 | 27 | 32.1421 | 147.882 | 3 | 0 | 1 |
| 11 | 50 | 63 | 113.083 | 304.078 | 2 | 5 | 5 |
| 12 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |


877_wt_z073c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 29992)
count
width (px)